Risk Management and Macroeconomic Disruptions in Supply Chains: The Role of Blockchain, Digital Twins, Generative AI, and Quantum Computing

Jiho Yoon, Rami Alkhudary, Srinivas Talluri, and Pierre Fénièss

***Abstract*—The global economy faces increasing vulnerabilities from macroeconomic disruptions, such as regulatory changes, trade tensions, geopolitical conflicts, currency volatility, pandemics, and energy crises that undermine the resilience of operations and supply chain management (OSCM) systems. These disruptions exacerbate risks, including supply chain breakdowns, operational inefficiencies, and systemic weaknesses, with energy challenges emerging as a key concern due to their effects on production costs, inflation, and sustainability goals. Advanced technologies, such as blockchain, digital twins, generative artificial intelligence (AI), and quantum computing, offer transformative potential to enhance transparency, predictive accuracy, and decision-making agility. However, their adoption introduces inherent tradeoffs, as they can lead to energy-intensive operations, cybersecurity risks, and economic burdens. To make sense of these dynamics, this article develops a conceptual framework based on a multilayered information system architecture that links specific disruptions to corresponding digital responses. This framework is grounded in a thorough review of both conceptual and empirical literature, along with extensive discussions among the authors. It explores how these technologies can address the risks stemming from macroeconomic disruptions while also considering their broader economic implications and challenges. It argues that simplistic solutions fail to account for the duality of these technologies' impacts and highlights the need for a systemic approach to integrate these technologies within OSCM. This article concludes by proposing actionable research directions for OSCM scholars and managers to navigate these complexities.**

***Managerial Relevance Statement*—This article offers practical guidance to operations and supply chain management professionals, engineering managers, and policymakers on mitigating risks associated with macroeconomic disruptions, including trade tensions, regulatory shifts, geopolitical conflicts, pandemics, and energy crises. It proposes the adoption of a next-generation information system based on a multilayered architecture that integrates blockchain, digital twins, generative artificial intelligence (AI), and quantum computing. For managers, blockchain can enhance data governance and integrity in cross-border transactions, thereby improving traceability and transparency. Digital twins support scenario simulations and virtual experimentation, helping managers anticipate disruptions and test contingency plans without interrupting operations. Generative AI processes unstructured data to generate predictive insights and dynamic recommendations, supporting informed decision-making in volatile conditions. Quantum computing addresses complex optimization problems under constraints, enabling managers to improve network design, inventory planning, and resource allocation. Moreover, our conceptualization helps both engineering managers and policymakers better understand new concepts such as autonomous, decentralized, and intelligent supply chains, as well as *omnitrust*, which refers to the potential replacement of interpersonal and interorganizational trust. This article also identifies key tradeoffs, including cybersecurity risks, sustainability concerns, and economic burdens. It presents clear propositions and recommendations, outlining the main challenges and the required groundwork for technology adoption.**

***Index Terms*—Blockchain, digital twins, generative artificial intelligence (AI), information system, macroeconomic disruptions, quantum computing, risk management, supply chain, TEM forum, technology management.**

I. Introduction

GLOBAL supply chains have become increasingly vulnerable to external disruptions, with macroeconomic shocks such as regulatory shifts, pandemics, and energy crises significantly affecting operations and decision-making in operations and supply chain management (OSCM). These shocks refer to large-scale, external forces originating from economic, political, or institutional systems. Such disruptions have led to heightened operational costs, longer lead times, and increased uncertainty across global markets [1], [2]. Conventional risk management strategies, such as redundancy, multisourcing, and nearshoring, have been adopted to mitigate these uncertainties [3], [4]. However, as disruptions become more frequent and complex, traditional approaches alone are no longer sufficient to ensure supply chain resilience in a volatile global economy [2], [5]. For example, the COVID-19 pandemic revealed the limitations of conventional risk management strategies, as organizations struggled with sudden demand fluctuations and supply chain bottlenecks [6]. To adapt to these emerging challenges, organizations must explore new solutions that offer enhanced adaptability and predictive capabilities.

One of the most promising approaches to strengthening supply chain resilience is the adoption of emerging digital technologies, including blockchain, digital twins, generative artificial

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Jiho Yoon is with the School of Business Administration, Chung-Ang University, Seoul 06974, South Korea (e-mail: jihoyoon@cau.ac.kr).

Rami Alkhudary and Pierre Fénièss are with Panthéon-Assas Research Lab in Management Science (LARGEPA), Paris Panthéon-Assas University, 75005 Paris, France (e-mail: rami.alkhudary@u-paris2.fr).

Srinivas Talluri is with the Department of Supply Chain Management, Michigan State University, East Lansing, MI 48824 USA (e-mail: talluri@broad.msu.edu).

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intelligence (AI), and quantum computing. These technologies provide innovative ways to address modern supply chain challenges by enhancing transparency, optimizing operations, and enabling data-driven decision-making [7], [8]. For example, blockchain facilitates real-time traceability of goods and automates regulatory compliance through smart contracts, which is useful in an era of heightened trade barriers and regulatory scrutiny [9]. In 2024, global spending on blockchain solutions was projected to reach 20.16 billion dollars. Forecasts indicate that this spending will continue to grow, reaching nearly 392.42 billion dollars by 2032 [10]. Moreover, recent studies highlight blockchain's role in ensuring supply chain sustainability by reducing inefficiencies and transaction costs [11].

Digital twins allow organizations to simulate potential disruptions and test alternative logistics strategies before they occur, offering proactive solutions to geopolitical risks and supply chain bottlenecks [2], [12]. Industries such as manufacturing, automotive, aviation, energy, utilities, healthcare, logistics, and retail have adopted digital twins to enhance productivity and operational efficiency. By 2025, the digital twin market in the manufacturing sector alone is projected to exceed 6 billion dollars [13]. Notably, during the COVID-19 pandemic, digital twin models were utilized to evaluate different supply chain configurations and optimize logistics in real time, demonstrating their effectiveness in mitigating pandemic-induced disruptions [12].

Generative AI improves demand forecasting and production planning, helping organizations adapt to fluctuating market conditions [14], [15]. Generative AI market is expected to grow rapidly, rising from 11 billion dollars in 2020 to nearly 128 billion dollars in 2024, and surpassing 1.3 trillion dollars by 2032. This growth is driven by the surge in generative AI tools in recent years, such as Bard by Google, ChatGPT by OpenAI, and Midjourney by Midjourney, Inc. [16]. Research indicates that AI-driven forecasting models outperform traditional statistical methods in handling volatile market conditions [17]. In addition, AI-powered automation enhances supply chain efficiency by reducing human error and expediting decision-making processes [18].

Quantum computing, though still in its early stages, has the potential to revolutionize complex scenario modeling and supply chain optimization [19]. The enterprise quantum computing market is expected to grow steadily between 2017 and 2030. In 2019 alone, global revenue from this market was projected to reach 167 million dollars [20]. Recent advancements suggest that quantum algorithms can enhance route optimization, risk assessment, and demand-supply balancing in global supply chains [21].

Despite their potential, these technologies introduce new challenges that organizations must carefully address. High energy consumption, cybersecurity risks, and concerns about workforce displacement present tradeoffs that require strategic consideration. The widespread adoption of AI and automated systems that could be powered by digital twins, for example, could lead to labor market disruptions, as automation reduces the demand for human labor in manufacturing and logistics [14]. Quantum computing also introduces potential security threats, as it has the capability to break conventional encryption methods, necessitating the development of quantum-resistant cryptographic protocols [22]. Meanwhile, blockchain's reliance on energy-intensive proof-of-work (PoW) mechanisms raises sustainability concerns, a particularly pressing issue in the context of global energy crises [23], [24]. However, alternative consensus mechanisms, such as proof-of-stake (PoS), have been proposed to reduce energy consumption [25]. In addition, a major security concern with public blockchains using PoW is the risk of centralization. For example, in 2021, the world's leading Bitcoin mining pools were all based in China, with five pools accounting for over half of the network's total hash rate, potentially undermining the blockchain's promise of decentralization [26].

Given these complexities, a critical question arises: how do macroeconomic disruptions drive the adoption of emerging technologies in OSCM? Furthermore, what specific mechanisms allow these technologies to enhance supply chain resilience, and what tradeoffs must organizations manage when implementing them? While existing research has explored the benefits of digital transformation, there remains a gap in understanding how systemic approaches can integrate multiple technologies to create robust, adaptive, and resilient supply chains.

This article aims to bridge this gap by analyzing the interplay between macroeconomic disruptions, digital technology adoption, and the systemic tradeoffs involved. By synthesizing insights from technology adoption models, supply chain risk management, and empirical case studies, this study contributes to the discourse on digital transformation in OSCM. We propose a conceptual framework that considers the interdependencies between macroeconomic forces, technological capabilities, and sustainability concerns in building long-term supply chain resilience. This framework will incorporate both academic and industry perspectives, ensuring that it is not only theoretically robust but also practically relevant for decision-makers navigating real-world supply chain challenges.

II. Literature Review

A. *OSCM Transformation and Macroeconomic Disruptions*

Logistics became central to the reconstruction of war-damaged economies following the Second World War. Its main focus was the optimization of transportation and inventory systems in a production context dominated by mass manufacturing and assembly-line methods, where supply often exceeded demand [27]. Countries needed efficient logistical networks to restore industrial capacity, ensure the flow of raw materials, and support distribution. In this phase, logistics was considered an operational function. During the 1980s, after the end of the Thirty Glorious Years, logistics expanded beyond transport and warehousing and began integrating with functions such as human resources, finance, and marketing. Michael Porter's value chain framework reflected this evolution, showing how logistics contributed to the overall value creation process by linking with other organizational activities [28].

In the 1990s, the spread of the internet and enterprise information systems enabled organizations to coordinate across organizational boundaries, transforming logistics into a strategic

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function [29]. Supply chain management emerged to describe this broader and more integrated role [30]. With real-time communication and global connectivity, logistics facilitated coordination among suppliers, manufacturers, and distributors. The globalization of supply networks was supported by this technological infrastructure. As a result, OSCM became a driver of competitive advantage in increasingly open markets, particularly in industries with complex and time-sensitive logistics requirements such as electronics, automotive, and consumer goods [31].

As has always been the case, but even more so today, OSCM faces growing pressure from geopolitical tensions, trade disputes, regulatory shifts, pandemics, and energy disruptions. Global economic policy uncertainty has surged in response to crises such as the 2001 dot-com bubble, the 2008 global financial crisis, and the 2020 COVID-19 pandemic. In response, two distinct strategic directions have emerged. One is the reorientation of supply chains toward more regional or local models, often referred to as partial deglobalization. This shift also reflects a transition from the just-in-time model to a more cautious just-in-case approach [32]. The second direction is driven by the rapid adoption of advanced technologies, such as blockchain, digital twins, generative AI, and quantum computing, among others.

These technologies are pushing OSCM toward systems that are decentralized and intelligent, representing a preliminary phase in the development of autonomous or more human-centric supply chains. This evolution aligns with the ambitions of Industry 5.0 or Society 5.0, and the emerging AI paradigm shift toward 6.0 generation systems [33]. The balance between these two paths will shape how supply chains approach risk management amid ongoing macroeconomic disruptions. The focus of our article is on the technological path.

B. Risk Management Amid Macroeconomic Disruptions

The increasing complexity of global supply chains has heightened organizations' exposure to macroeconomic risks, necessitating adaptive risk management strategies. Supply chain risks, as defined by Christopher [34], encompass any disruptions to the smooth flow of goods and services. Ivanov [35] further categorizes macroeconomic disruptions, such as pandemics, financial crises, trade conflicts, and regulatory shifts, as key drivers necessitating strategic adjustments.

Macroeconomic risks can be categorized into political, economic, and environmental dimensions, each imposing distinct challenges on OSCM (see Fig. 1). Political risks, including trade tensions, international conflicts, and domestic instability, alter trade flows and increase transaction costs. For example, escalating trade tensions between major economies have prompted organizations to reevaluate sourcing strategies in response to shifting regulatory barriers [36], while geopolitical instability, such as the Russia-Ukraine war, has contributed to energy crises, raising production costs and disrupting logistics [37], [38]. Economic risks, such as exchange rate fluctuations and financial downturns, directly affect procurement costs, inventory valuation, and demand volatility [39]. The COVID-19 pandemic exemplifies an environmental disruption that severely

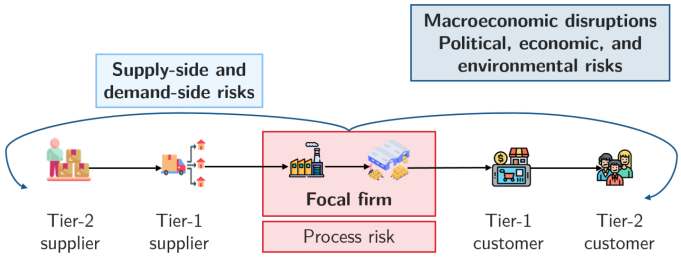


Fig. 1. Expanding risk sources in OSCM, based on the framework of Christopher and Peck [42]. (Icon credits: flaticon.com).

destabilized global logistics and production, increasing lead time uncertainties and inventory shortages [6], [40]. Similarly, regulatory measures such as the European Union's Carbon Border Adjustment Mechanism have imposed sustainability compliance costs, altering global production and distribution networks [41]. These examples underscore the systemic impact of macroeconomic uncertainties on supply chain stability.

Traditionally, organizations have employed resilience strategies such as supplier diversification, inventory buffering, and flexible contracting to mitigate these risks [3], [5]. While effective in some contexts, these approaches often fail to address the scale, speed, and unpredictability of contemporary macroeconomic shocks. First, many resilience strategies rely on historical data and static contingency plans, which become inadequate in volatile and rapidly evolving crisis scenarios [35], [43]. Second, excessive inventory buffering increases holding costs and financial risk in industries with fluctuating demand [1]. Third, supplier diversification can introduce complexity and coordination challenges, reducing visibility and agility rather than enhancing them [34]. Furthermore, globalized supply networks have increased interdependencies, meaning disruptions propagate more widely and unpredictably than traditional risk models anticipate [6], [44].

Given these challenges, supply chain resilience now demands real-time adaptability, predictive risk assessment, and end-to-end visibility [1], [18]. Static, reactionary strategies are insufficient in addressing dynamic macroeconomic disruptions, necessitating a more data-driven, interconnected, and proactive approach. The following section examines how emerging technologies align with these requirements, enabling organizations to anticipate, assess, and respond to macroeconomic shocks with greater precision and agility.

To address the challenges posed by macroeconomic disruptions, businesses have increasingly turned to advanced technologies as a means of enhancing resilience, efficiency, and transparency within their supply chains [45]. The existing body of research highlights the advantages that technologies such as blockchain, generative AI, digital twins, and quantum computing can offer in the realm of supply chain risk management [46], [47], [48], [49], [50], [51]

III. Technologies and Their Applications

The growing complexity and uncertainty within OSCM highlight the need for advanced technologies to mitigate risks

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TABLE I
TECHNOLOGY APPLICATIONS ACROSS MACROECONOMIC DISRUPTORS

Technology	Trade Tensions	Regulatory Changes	Pandemics	Energy Issues	Technology Maturity	Source
Blockchain	Real-time regulation tracking	Smart contract compliance	Supply traceability	Peer-to-peer energy trading	Adoption underway	Deloitte (2023) [52]
Digital Twins	Logistics rerouting	Emissions optimization	Production simulation	Grid load balancing	Adoption underway	Gartner (2024) [53]
Generative AI	Demand forecasting	Design compliance	Supply chain reconfiguration	Energy efficiency	Emerging	McKinsey (2023) [54]
Quantum Computing	Network optimization	Compliance forecasting	Logistics optimization	Price fluctuation prediction	Nascent	IBM (2024) [55]

arising from macroeconomic disruptions. This section explores the applications of four key technologies, namely blockchain, generative AI, digital twins, and quantum computing. We aim to articulate how these emerging technologies can interact and complement one another within OSCM, forming an integrated ecosystem that supports the development of more autonomous, decentralized, and intelligent supply chains. To synthesize their practical roles, Table I summarizes how blockchain, digital twins, generative AI, and quantum computing address key macroeconomic disruptions, including trade tensions, regulatory changes, pandemics, and energy issues.

A. *Blockchain: Data Management Layer*

Blockchain technology has emerged as a transformative tool for enhancing transparency and traceability within supply chains [56], particularly in response to geopolitical conflicts and regulatory shifts. By providing a decentralized and immutable ledger, blockchain ensures the integrity of transactional data, mitigating risks such as counterfeiting and fraud [57], [58]. For example, during the Russia-Ukraine conflict, blockchain was used to secure and trace critical goods, ensuring that supplies such as pharmaceuticals were not tampered with [59]. In the context of pandemics, the blockchain improves traceability, as evidenced by Walmart's 2020 initiative to track food supplies from farm to store in seconds during the COVID-19 pandemic, helping to maintain safety and availability despite disruptions in logistics networks [60]. Beyond its security advantages, blockchain also facilitates more efficient cross-border transactions, which proves valuable when trade regulations change or become unpredictable [61]. This ability to maintain transaction integrity and streamline cross-border exchanges makes the blockchain valuable in an increasingly volatile global trade environment.

To fully realize the benefits of blockchain in OSCM, organizations must undertake a fundamental transformation of their business models. This transformation involves adopting new routines and operational processes that support the seamless integration of information systems [62]. A critical aspect of this transformation is the handling of metadata associated with both physical and financial flows within the supply chain. For blockchain systems to operate in a truly decentralized manner, this metadata must be stored on distributed cloud infrastructures [63]. This applies to both public and private blockchain configurations. In most cases, only cryptographic hashes of the metadata are recorded on the blockchain ledger, while the metadata itself is stored elsewhere. If that storage takes place on the cloud or centralized servers, the system risks becoming centralized [64].

In addition, ensuring that the information recorded on the blockchain ledger reflects physical reality is a critical requirement for trustworthy supply chain applications. This challenge, often discussed in the context of the physical internet [65]. Achieving this alignment requires minimizing human intervention in the data capture process, which is prone to errors, delays, and manipulation. One approach is the integration of Internet of Things (IoT) devices across the supply chain. These devices, including sensors, scanners, and RFID tags, can detect, collect, and transmit real time data related to location, temperature, movement, and condition of goods. The integration of IoT with blockchain reduces the risk of human error and increases confidence in the integrity of supply chain data [66].

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 1: Blockchain mitigates risks related to macroeconomic disruptions by ensuring the integrity and auditability of transactional records in cross-border supply chains, which supports operational continuity when trust in centralized institutions or financial intermediaries is compromised.

B. *Generative AI: Analytics and Decision Support Layer*

Generative AI plays a key role in enhancing the efficiency and resilience of supply chains. Its applications are wide-ranging, from demand forecasting and predictive maintenance to risk management. It can help mitigate risks posed by disruptions such as pandemics, including COVID-19, by optimizing inventory management and demand planning [67]. With its ability to process large datasets quickly, generative AI is also valuable in identifying potential disruptions triggered by geopolitical events, such as trade tensions and changes in regulatory environments [68].

As global trade and market dynamics become increasingly volatile, generative AI's capacity to forecast and manage supply chain risks is becoming more critical. During the COVID-19 pandemic, generative AI-powered platforms enabled businesses to optimize operations and mitigate the impacts of supply chain disruptions [67]. It also enhances decision-making by facilitating real-time analysis of market trends and suggesting strategies to minimize exposure to risks. For example, during periods of heightened trade tensions, generative AI can predict demand shifts. In 2019, Amazon used AI-driven forecasting to adjust

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inventory amidst global trade uncertainty, reducing excess stock by 12% and mitigating cost increases [54]. In response to regulatory changes, it can design systems that ensure compliance; Tesla's use of generative AI to optimize battery production under EU emissions standards reduced material waste by 25%, aligning with green deal targets [54]. In energy contexts, generative AI contributes to efficiency; oil refineries applied it during the 2022 price surges to optimize processes, saving 10% in energy costs amidst soaring gas prices [54].

Generative AI holds a lot of potential to address Herbert Simon's concept of bounded rationality, which highlights the limitations individuals face in decision making due to constraints in time, information availability, and cognitive capacity [69]. Generative AI models can mitigate these constraints by delivering real time data analysis, generating context-specific insights, and offering adaptive recommendations, thereby enabling managers to make more informed and optimized decisions under uncertainty [70]. Unlike traditional decision support tools, generative AI can synthesize complex inputs, simulate scenarios, and propose actionable options, expanding the decision maker's rational boundary.

However, beyond its functional capabilities, a critical question emerges regarding the role generative AI will play within OSCM. Precisely, will it be adopted as an add-on to existing information systems, or will it drive deeper transformation in business models, operational processes, and organizational routines? While this question is common with the introduction of new technologies, the case of generative AI may be different due to the scale of investment it attracts and its wide-ranging impact, not only in fields such as education but also in how supply chains are managed and decisions are structured. Its influence can range from tactical improvements to strategic shifts, making it a turning point for organizations.

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 2: Generative AI mitigates risks related to macroeconomic disruptions by synthesizing large-scale, unstructured data to generate predictive insights and adaptive strategies, enabling organizations to proactively adjust to volatile market conditions and shifting demand patterns.

C. Digital Twins: Simulation and Representation Layer

Digital twin technology is gaining popularity in OSCM due to its ability to simulate and monitor real-time operations. By creating a virtual replica of the physical supply chain, digital twins allow businesses to track the status of goods, production processes, and logistics in real time. This technology helps organizations simulate potential disruptions caused by external factors, such as regulatory changes or geopolitical conflicts, and adjust their strategies accordingly [71]. For example, during the pandemic, companies used digital twins to simulate the impacts of lockdowns and transportation restrictions, enabling them to make real-time adjustments to their plans [72].

In addition, digital twin technology mitigates risks by offering businesses a comprehensive understanding of their supply chain dynamics. This insight is valuable in the context of external disruptions, such as trade tensions, as it enables companies to identify vulnerabilities and take proactive measures to minimize potential disruptions [73]. During physical conflicts, such as the Russia–Ukraine war, digital twins model logistics network resilience; Siemens utilized them in 2022 to reroute shipments around damaged Ukrainian infrastructure, reducing transit delays by 10% and ensuring continuity [53]. For pandemics, they simulate production capacity under constrained conditions, as GE Aviation did during COVID-19 to adjust jet engine assembly lines, sustaining output despite a 20% workforce reduction caused by illness and lockdowns [53]. Addressing regulatory changes, digital twins optimize compliance with emissions standards; manufacturers piloting twins under the EU's green deal reduced carbon footprints by up to 15% by simulating energy-efficient processes, adapting to new mandates [53].

Digital twins enable organizations to sense changes in their operations or environment, seize emerging opportunities, and transform processes in response to evolving conditions, a process that can be better understood through the lens of dynamic capabilities [74]. By creating a virtual representation of physical systems that is continuously updated with real time data, digital twins provide a powerful tool for monitoring performance and testing improvements without disrupting actual workflows. This capability supports lean principles by offering end to end visibility and enabling managers to identify inefficiencies, eliminate waste known as muda, and optimize operations [75]. Through simulation and data driven analysis, managers can explore solutions such as adjusting production schedules, rebalancing inventory, or reallocating resources. These actions foster a continuous improvement cycle rooted in empirical evidence and aligned with methodologies like Kaizen or the Plan Do Check Act approach. In this way, digital twins serve as a strategic resource that helps organizations remain responsive in complex supply chain environments.

Drawing on this reasoning and internal discussions among the co-authors of this work, we propose the following.

Proposition 3: Digital twins mitigate risks related to macroeconomic disruptions by creating continuously updated virtual replicas of supply chain systems, allowing organizations to test contingency plans, identify bottlenecks, and coordinate adjustments without interrupting physical operations.

D. Quantum Computing: Computational Optimization Layer

Quantum computing has the potential to advance supply chain optimization. Unlike classical computing systems that process data in binary form, quantum computers utilize quantum bits, enabling them to process information at an exponentially faster rate. This enhanced computational capability is valuable in addressing complex optimization problems. In the context of OSCM, quantum computing can improve decision-making processes in areas such as inventory management, route optimization, and demand forecasting by analyzing and processing large volumes of data [57]. For trade disputes, it optimizes supply chain networks under regulatory constraints; Volkswagen's 2022 simulations rerouted logistics amid trade tensions, reducing

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costs by 8% [55]. Addressing regulatory changes, quantum algorithms forecast compliance scenarios; organizations modeling carbon pricing under the green deal gained strategic insights into cost impacts [55]. During pandemics, it enhances logistics optimization; D-Wave's 2020 pilot with shipping organizations cut port congestion delays by 15% during COVID-19 [55]. In energy markets, it improves forecasting; during the 2022 crisis, it predicted price fluctuations with 20% greater accuracy, aiding resource planning [55].

Although still in the early stages of development, quantum computing demonstrates considerable potential for enhancing supply chain operations. Its applications are pertinent when addressing disruptions, such as geopolitical conflicts, where rapid and precise decisions regarding routing and inventory management are crucial. Preliminary research indicates that quantum computing could enhance the responsiveness and flexibility of supply chains in mitigating the impacts of macroeconomic disruptions [76].

The potential impact of quantum computing is often framed through the lens of computational complexity theory [77], [78]. Quantum computers operate within the complexity class known as bounded error quantum polynomial time. While it remains an open question whether quantum computing can solve nondeterministic polynomial time-complete problems, certain specific tasks are already known to be accelerated by quantum algorithms. One prominent example is Shor's algorithm, which transformed the problem of integer factorization from one requiring exponential time on classical machines to one solvable in polynomial time on a quantum computer.

As summarized by the Harvard Business Review [79], this breakthrough reduced the time required to factor large numbers from billions of years to just a few days. This shift has major implications for OSCM. For example, quantum heuristics could enable inventory optimization over exponentially large decision spaces that are currently beyond reach. Similarly, Grover's algorithm offers a quadratic speed-up for unstructured search problems, which could accelerate tasks such as scanning massive datasets or exploring complex decision models. However, a study by Boston Consulting Group [80] highlighted that, at present, quantum computing offers no practical superiority over classical systems in most commercial applications, though researchers agree that quantum advantage is likely to emerge in specific problem areas where classical methods cannot compete [81], [82].

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 4: Quantum computing mitigates risks related to macroeconomic disruptions by solving complex optimization problems involving constrained resources and network interdependencies, enhancing long-term planning capabilities in scenarios of geopolitical conflict, regulatory change, or systemic shocks.

E. Toward a Next-Generation Information System

The integration of blockchain, digital twins, generative AI, and quantum computing within OSCM can be described as a multilayered architecture (see Fig. 2). At the foundational data management layer, IoT devices continuously capture real-time operational data, such as location, status, condition, and movement of physical assets, providing accurate, high-frequency streams while reducing human error. When governed by blockchain, this layer can also incorporate inputs related to macroeconomic disruptions from various entities across the supply chain, with cryptographic hashes of metadata stored securely and transparently. The second layer, known as the simulation and representation layer, involves digital twins, which use data from IoT and blockchain to construct virtual replicas of physical supply chain operations, enabling scenario simulations, condition monitoring, and virtual experimentation. Above this is the analytics and decision support layer, where generative AI models understand unstructured sources and enhance the integrated data to produce predictive insights, dynamic recommendations, and actionable strategies, thus supporting complex decision-making processes. Finally, at the topmost computational optimization layer, quantum computing is employed to resolve complex optimization problems involving vast datasets or extensive combinational analyses, surpassing the computational capabilities of classical computing systems.

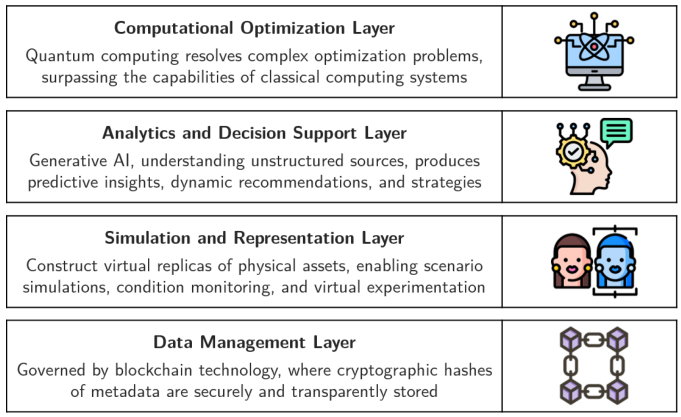


Fig. 2. Multilayered architecture integrating blockchain, digital twins, generative AI, and quantum computing in OSCM. (Icon credits: flaticon.com).

Together, these integrated layers could establish a robust technological architecture capable of supporting autonomous, decentralized, and intelligent supply chain information systems. The convergence of these technologies could also introduce the concept of *omnitrust*, whereby a technological ecosystem could replace traditional reliance on interpersonal or interorganizational trust. Such a system reduces human biases and informational asymmetries, decreasing dependence on subjective trust. In an omnitrust-enabled supply chain, decisions traditionally dependent on human judgment or interorganizational negotiation are instead executed autonomously, guided by smart contracts, algorithmically verified insights and context-specific recommendations. By incorporating analytics and simulations of macroeconomic disruptions, the system proactively addresses emerging risks. Consequently, the adoption of omnitrust shifts supply chain decision-making from reactive and trust-based practices to proactive, adaptive, and autonomous operations, enhancing organizational responsiveness and resilience in dynamic environments.

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The adoption of a multilayered architecture integrating blockchain, digital twins, generative AI, and quantum computing in OSCM represents a complex transformation that extends beyond technical implementation. To understand and guide this transition, established theoretical frameworks offer complementary perspectives. For example, the technology-organization-environment framework is well suited for analyzing the broader context of adoption by considering technological readiness, organizational capabilities, and external environmental pressures [83]. The technology acceptance model (TAM) focuses on individual users' perceptions of usefulness and ease of use, which are essential for the successful deployment of system interfaces that require human interaction and trust in system outputs [84]. The Unified Theory of Acceptance and Use of Technology (UTAUT) extends TAM by incorporating social influence and facilitating conditions, making it valuable for understanding collective adoption behavior within and across organizations [85]. Finally, the Technology Readiness Index provides insight into the psychological and behavioral predispositions of organizations or managers toward embracing such systems [86].

In line with the adoption of this multilayered architecture integrating blockchain, digital twins, generative AI, and quantum computing, a critical prerequisite lies in the domain of data engineering [87], [88], [89]. To enable seamless interaction between these technologies, particularly at the blockchain-governed data management layer, it is essential to merge and standardize structured data (such as transactional logs and inventory records) with unstructured data (such as sensor outputs, scanned documents, and shipment narratives) sourced from various systems across the supply chain. This integration requires careful data cleansing, formatting, and transformation to ensure consistency, compatibility, and integrity.

In addition, an important complement to these technological advancements is the adoption of standards such as ISO 22301 [90], which provides a structured framework for Business Continuity Management Systems. ISO 22301 enhances organizational awareness of systemic risks, including those triggered by macroeconomic disruptions such as inflation, trade imbalances, and monetary policy shifts. By guiding organizations in identifying critical functions, assessing impacts, and designing recovery strategies, the standard supports proactive risk management. ISO 22301 enables continuous monitoring and coordinated response mechanisms, aligning with the data-driven, adaptive capabilities of digital infrastructures. This integration strengthens supply chain resilience by embedding continuity planning into operational and strategic layers of decision-making.

Lastly, another major challenge is achieving interoperability across heterogeneous systems and data formats [91], [92], as supply chain actors often operate on incompatible platforms and standards. Addressing this requires the development and adoption of shared protocols, standardized data schemas, and integration interfaces to enable seamless data exchange across organizational and technological boundaries [93].

IV. Macroeconomic Implications and Challenges

Blockchain technology has demonstrated its capacity to enhance traceability and mitigate the risk of fraud, thus improving overall supply chain visibility and security [15], [58], [94] AI-driven analytics, on the other hand, have shown promise in enhancing demand forecasting and inventory optimization, enabling businesses to better respond to fluctuations in market conditions [15], [95]. For example, Amazon employs AI-driven systems to optimize inventory management, balancing stock levels to prevent both surpluses and shortages. These systems have reduced order fulfillment costs and are expected to save the company 10 billion dollars annually by 2030 [96]. Digital twins, a virtual representation of physical assets, facilitate real-time monitoring and simulation of supply chain operations, allowing for more proactive risk management [97], [98] In addition, quantum computing has emerged as a powerful tool for solving complex optimization problems, enabling businesses to simulate and predict risk scenarios in ways that traditional computing methods cannot [21], [99].

These technological advancements have been applied in real-world OSCM scenarios with notable success in the era of macroeconomic disruptions [100]. While the adoption of these advanced technologies has yielded numerous benefits, it is not without its challenges and potential negative consequences. The energy-intensive nature of blockchain and AI technologies could raise sustainability concerns, and furthermore, AI-driven automated systems that could be powered by digital twins can also disrupt labor markets, potentially leading to workforce displacement [15], [100], [101], [102], [103]. In addition, the emergence of quantum computing poses a threat to data security and existing encryption standards [15].

While these issues vary depending on the specific technology and its application, three key concerns stand out: energy consumption, security risks, and economic burdens. These concerns highlight the dual nature of technology adoption, where the benefits often come with unintended consequences. The economic tradeoffs associated with these technologies across different disruption types are summarized in Table II, where solutions to one problem may generate new challenges, revealing the dual nature of technology in OSCM, This illustrates how each innovation, while addressing specific challenges, can introduce new risks.

A. Security Risks: Safeguarding Data and Privacy

As supply chains become increasingly digitalized, the risks related to data security and privacy have grown substantially. Blockchain, while known for its ability to enhance transparency and traceability, also raises concerns about the protection of sensitive data. The decentralization inherent in blockchain systems can expose supply chains to cyberattacks of the metadata associated with the hashes on the blockchain ledger, if proper safeguards and decentralized storage are not implemented [62], [63]. Moreover, AI technologies can be vulnerable to adversarial attacks that manipulate algorithms or compromise data integrity [59]. These security concerns are critical, as any breach could disrupt operations, undermine consumer trust, and expose companies to financial and reputational risks [68].

Furthermore, as digital twin technology enables real-time monitoring and simulation of supply chains, the accumulation of

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TABLE II
MACROECONOMIC IMPLICATIONS OF TECHNOLOGIES ACROSS DISRUPTORS

Technology	Trade Tensions	Regulatory Changes	Pandemics	Energy Issues	Source
Blockchain	Cost savings versus energy costs	Compliance gains versus scale	Resilience versus energy strain	Innovation versus grid strain	Deloitte (2023) [52]
Digital Twins	Efficiency versus data risks	Savings versus compliance costs	Output versus energy reliance	Stability versus power demands	Gartner (2024) [53]
Generative AI	Forecasting versus labor risks	Efficiency versus energy use	Agility versus job displacement	Savings versus training costs	McKinsey (2023) [54]
Quantum Computing	Optimization versus security	Innovation versus compliance	Speed versus cybersecurity	Precision versus cooling costs	IBM (2024) [55]

vast amounts of data raises issues regarding the storage and protection of sensitive information [104]. The more interconnected the systems, the higher the likelihood of potential data breaches or unauthorized access. Similarly, quantum computing, which promises to revolutionize optimization in supply chains, presents a unique challenge in terms of data encryption and the safeguarding of proprietary information [57]. The rapid advancement of these technologies demands that robust security measures be integrated into their design and deployment to mitigate potential threats.

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 5: Blockchain, generative AI, digital twins, and quantum computing introduce security risks, including metadata leakage, model manipulation, real-time data breaches, and cryptographic vulnerability, requiring technology-specific protection measures.

B. Economic Burden: Costs of Adoption and Maintenance

The economic burden associated with the adoption and maintenance of advanced technologies is another key challenge. Implementing technologies like blockchain, generative AI, digital twins, and quantum computing often requires upfront investment in infrastructure, training, and system integration. These costs can be burdensome for small and medium-sized enterprises (SMEs), which may struggle to afford the capital needed for technology adoption [61]. In addition, the maintenance of these systems, including regular updates, troubleshooting, and energy consumption, adds to the ongoing financial strain.

For example, quantum computing, despite its potential for solving complex optimization problems, remains a costly investment with limited availability [76]. As such, only large corporations with substantial resources can afford to explore its potential, creating a technological divide between large and small organizations. Similarly, the integration of AI systems, blockchain and digital twins requires constant data input and tuning, which can incur significant operational costs over time. These economic barriers must be addressed through financial incentives, subsidies, or collaborative partnerships to ensure that the benefits of these technologies are accessible to a broader range of businesses.

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 6: Blockchain, generative AI, digital twins, and quantum computing demand high investment in infrastructure, expertise, and integration, posing cost barriers for SMEs and reinforcing inequality in supply chain digitalization.

C. Energy Consumption and Sustainability Concerns

The rapid deployment of advanced technologies often requires substantial computational power, which translates into higher energy usage. For example, AI and quantum computing, due to their data-intensive nature, have been found to demand enormous amounts of electricity for training algorithms and running simulations, leading to an increase in carbon footprints [72].

The energy-intensive nature of these technologies becomes especially concerning when considered against the backdrop of global efforts to reduce carbon emissions and transition to more sustainable energy sources. Digital twins, while offering real-time monitoring and simulation, also require significant data processing power, raising concerns about their environmental impact, particularly if the underlying infrastructure is powered by nonrenewable energy sources [73]. Furthermore, blockchain, despite its security and transparency benefits, involves a decentralized system that often necessitates continuous, high-energy consumption, particularly in PoW consensus mechanisms [59]. Although the use of public blockchains and PoW is usually associated with financial transactions and smart contracts, these systems can also be used to register the hashes of metadata, serving as a universal source of truth whenever metadata are changed or modified [62].

The challenge, therefore, lies in balancing the potential of these technologies with the need for sustainability. As industries strive to adopt these technologies to combat disruptions, the environmental impact must be carefully managed to avoid exacerbating existing ecological issues. This brings to the forefront the necessity of a systemic approach, one that not only considers the technological capabilities but also integrates sustainability into the decision-making process.

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 7: Blockchain, generative AI, digital twins, and quantum computing increase energy consumption through intensive computation, highlighting the need for energy-efficient design and renewable power integration to reduce environmental impact.

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D. Need for a Systemic Approach

Given the dual nature of these technologies, offering benefits but also introducing risks, the need for a systemic approach becomes critical. This approach ensures that the implementation of advanced technologies takes into account not only their operational efficiency and risk mitigation potential but also their environmental, social, and economic impacts. A systemic approach involves designing strategies that integrate these technologies while addressing their energy consumption, sustainability, and security concerns. This holistic perspective is essential for mitigating the potential unintended consequences of rapid technological adoption.

As these technologies evolve and become more integrated into global supply chains, the focus must shift toward creating frameworks that balance innovation with responsibility. This will involve developing energy-efficient algorithms, utilizing renewable energy sources, and adopting practices that minimize environmental impact. The systemic approach, therefore, is not merely a strategic necessity but an ethical imperative for ensuring that the adoption of advanced technologies contributes to both supply chain resilience and global sustainability goals.

Drawing on this reasoning and internal discussions among the coauthors of this work, we propose the following.

Proposition 8: Blockchain, generative AI, digital twins, and quantum computing require a systemic approach that aligns technical deployment with sustainability, security, and cost considerations to ensure responsible and coordinated adoption in OSCM.

V. Conclusion

The integration of advanced technologies is reshaping the landscape of OSCM in response to macroeconomic disruptions. These disruptions, arising from political instability, economic volatility, regulatory shifts, or environmental crises, expose supply chains to systemic risks. To remain resilient, organizations are increasingly turning to a unified digital infrastructure that leverages real-time data, intelligent analysis, and advanced computational power to support faster, more adaptive, and more robust decision-making across the supply chain network.

Blockchain enhances transparency and transactional trust by recording verifiable data across decentralized systems, which is important when institutional reliability is in question. Digital twins enable real-time simulation of physical systems, allowing organizations to test contingency plans and adjust operations virtually before implementing physical changes. Generative AI synthesizes large, complex, and often unstructured data sources to generate insights and dynamic strategies, helping organizations adapt to fast-changing demand patterns and regulatory environments. Quantum computing extends this digital architecture by solving complex optimization problems involving resource constraints, network disruptions, or competing objectives, offering a level of computational performance that exceeds classical systems.

Despite these benefits, the adoption of blockchain, digital twins, generative AI, and quantum computing introduces many challenges. Security risks are a major concern, as each system creates new attack surfaces, ranging from data leakage in blockchain and real-time breaches in digital twins, to adversarial manipulation in AI and future vulnerabilities in encryption due to quantum capabilities. Economic burdens related to infrastructure investment, technical expertise, and system integration are particularly acute for SMEs, which may struggle to adopt and maintain these technologies. In addition, the energy-intensive nature of these systems raises sustainability concerns, especially when the computational load is not offset by renewable energy sources or efficient design. These challenges highlight the need for coordinated, system-wide strategies that balance innovation with responsible implementation.

In this article, we propose a conceptual framework based on a multilayered information system architecture that captures the interplay between macroeconomic disruptions, digital technology adoption, and systemic tradeoffs (see Fig. 3). At the foundation is the data management layer, governed by blockchain, which hashes and secures metadata from various points in the supply chain, including signals related to macroeconomic disruptions. The simulation and representation layer builds on this data using digital twins to construct dynamic, real-time models of supply chain operations, enabling virtual experimentation and continuous monitoring. The analytics and decision support layer incorporates generative AI to interpret unstructured data streams, identify patterns, and generate predictive, context-aware recommendations. At the top, the computational optimization layer employs quantum computing to solve high-complexity problems that require the rapid processing of large-scale, multidimensional data.

A. Recommendations and Future Research Directions

The adoption of blockchain, digital twins, generative AI, and quantum computing into OSCM systems offers a robust framework for mitigating the multifaceted risks posed by macroeconomic disruptors, trade tensions, regulatory changes, pandemics, and energy issues, as explored in preceding sections. These technologies, while transformative, introduce challenges such as energy consumption, cybersecurity vulnerabilities, and socioeconomic impacts that demand strategic responses to ensure economic stability. This section provides actionable recommendations for engineering and technology managers to employ their potential, addressing each disruptor's unique pressures, and outlines future research directions to advance OSCM scholarship. These recommendations aim to fortify supply chains against trade tensions, regulatory shifts, health crises, and energy volatility. Concurrently, the proposed research avenues seek to deepen understanding of these technologies' systemic interactions, offering a forward-looking perspective that complements their practical deployment without revisiting earlier analyses.

For engineering and technology managers, a primary recommendation is to prioritize energy-efficient solutions to counter the energy demands highlighted across these technologies, particularly in the context of energy issues that amplify operational costs and sustainability pressures. Ethereum's transition from PoW to PoS, completed in 2022, has resulted in a 99.95% reduction in energy consumption, making it roughly 2000 times more

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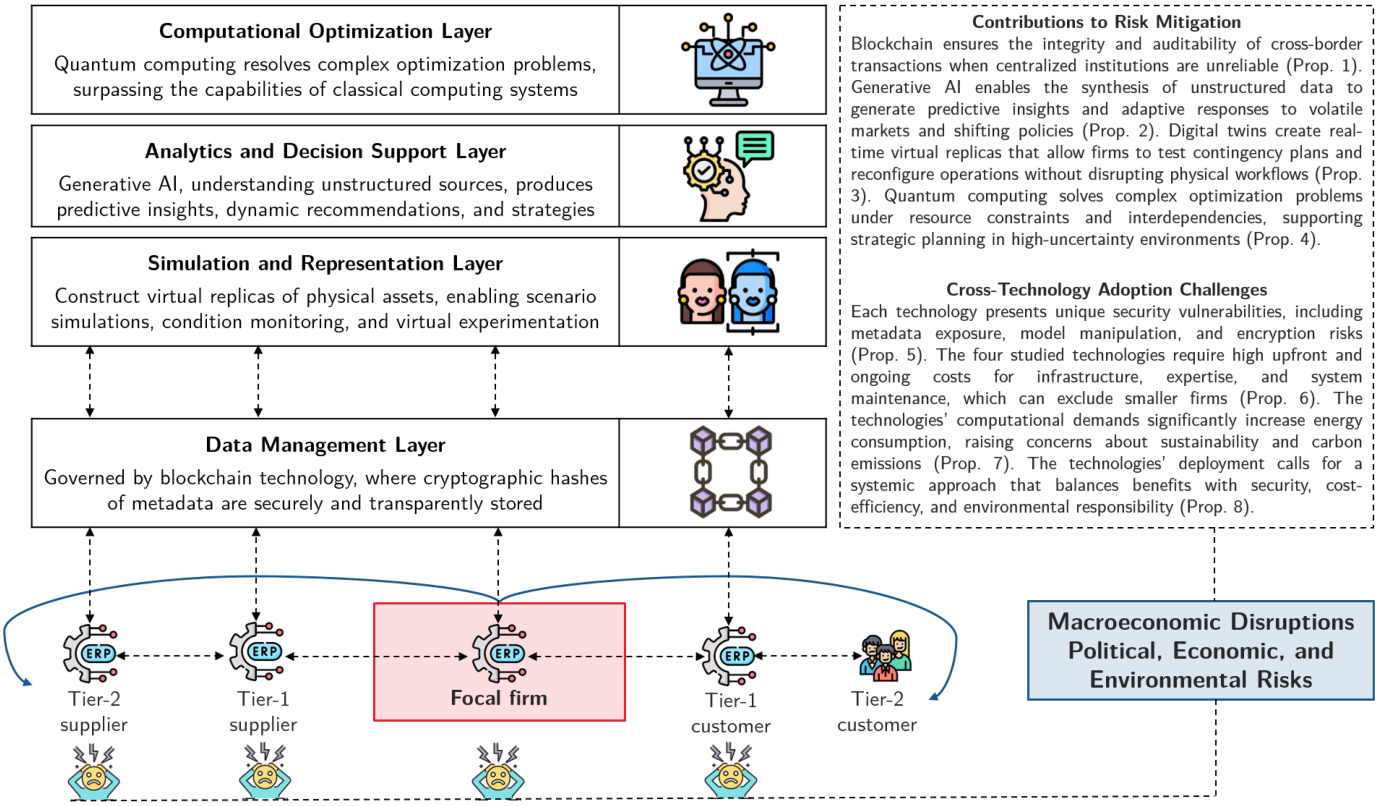


Fig. 3. Conceptual framework of contributions and challenges in using blockchain, digital twins, generative AI, and quantum computing to mitigate macroeconomic disruption risks in OSCM. (Icon credits: flaticon.com).

energy-efficient. This change led to a significant decrease in the carbon footprint of a single Ethereum transaction, from 109.71 kg to 0.01 kg, as confirmed by tests run on the Beacon Chain prior to the Merge [105]. Implementing this in supply chain tracking could save organizations 5–10 million dollars annually in energy costs during crises such as the 2022 European gas shortage, aligning with sustainability mandates [52]. Generative AI's training, requiring 1287 megawatt-hours per large model, can be mitigated by adopting smaller, task-specific models, or leveraging renewable energy sources; Google's carbon-neutral data centers offer a model, reducing AI's environmental impact by 40% [106]. Digital twins, with moderate consumption of 50–100 kWh monthly per factory, benefit from edge computing to localize processing, cutting cloud energy use by 20% and enhancing resilience during energy disruptions [53]. Quantum computing's cooling demands, at 100–200 kWh daily, require investment in hybrid quantum-classical systems, as IBM's 2023 pilots suggest a 30% energy reduction for optimization tasks [55]. These strategies ensure OSCM systems remain viable amidst energy volatility, supporting economic stability without compromising operational efficiency.

Beyond energy, managers must address trade tensions and regulatory changes through adaptive technology deployment to maintain competitiveness and compliance. Blockchain's smart contracts can automate regulatory adjustments, as piloted by Maersk in 2020 to manage international trade shifts, reducing compliance costs by 12% and enabling rapid sourcing pivots [52]. Digital twins model supply chain rerouting under trade disputes; Siemens' 2022 reconfiguration around Ukraine saved 8 million dollars in logistics costs. Generative AI's forecasting capabilities, used by Amazon to cut inventory costs by 12% during trade tensions, can predict regulatory impacts, enabling proactive design adjustments to meet standards such as the EU's Green Deal [54]. Quantum computing's scenario modeling, as Volkswagen's 2022 trials showed with an 8% cost reduction, supports strategic planning for trade and compliance shifts [55]. Integrating these tools into OSCM workflows ensures agility in navigating trade barriers and regulatory landscapes, minimizing disruptions while optimizing resource allocation.

In the face of macroeconomic disruptions, including geopolitical tensions, supply chain disruptions, and financial instability, robust data systems and workforce strategies are essential for maintaining supply chain resilience. Digital twins, which rely on real-time data to optimize operations, can provide valuable insights during such disruptions. However, they are vulnerable to failures in IoT networks, which can undermine system stability. For example, GE Aviation's 2020 investment of 15 million dollars in backup sensors ensured 90% production uptime, mitigating the impact of external shocks, but this required significant costs and resource investments [53]. Blockchain technology enhances traceability and transparency in supply chains, as demonstrated by Walmart's partnership with IBM. Before adopting blockchain, tracing a package of mangoes took over six days. However, with the new blockchain-based system, Walmart was able to trace products from source to store in just 2.2 s, improving efficiency and supporting supply continuity [60]. However, its energy consumption remains a challenge, requiring further efforts to improve sustainability.

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TABLE III
FUTURE RESEARCH DIRECTIONS ACROSS TECHNOLOGIES AND THEMES

Technology	Energy and Sustainability	Workforce Adaptation and Social Impacts	Governance and Cybersecurity	Systemic Interactions and Modeling	Knowledge Management Systems (KMS)	Omnitrust and Autonomous Supply Chains	Geopolitical Strategy and Territorial Dynamics
Blockchain	Explore PoS energy efficiency in supply chains	Assess workforce readiness and training needs for blockchain adoption	Develop quantum-safe blockchain protocols	Analyze blockchain's role in multi-tech disruption response	Design blockchain-enabled KMS for real-time traceability	Investigate governance of decentralized trustless networks	Assess blockchain in Arctic trade and territorial shifts
Digital Twins	Use edge computing to reduce energy in modeling	Identify skills for twin deployment	Improve digital twin cybersecurity frameworks	Simulate cascading disruption impacts across OSCM layers	Integrate simulation insights into KMS for decision support	Study autonomous twin-driven coordination	Model territorial effects on supply infrastructure
Generative AI	Study low-resource LLMs and green model training	Assess AI-driven displacement and reskilling	Establish AI governance frameworks	Evaluate AI for integrated planning and forecasting	Develop generative AI-based KMS for decision workflows	Examine role of agents in trustless coordination	Analyze AI for geopolitical and strategic realignment
Quantum Computing	Design hybrid quantum-classical models to offset infrastructure energy demands	Identify quantum skill gaps and develop specialized education programs	Define governance for quantum cybersecurity risks	Optimize across disruption scenarios using quantum models	Explore quantum-enhanced knowledge discovery	Explore autonomous quantum scheduling systems	Weigh sovereignty and access in quantum deployment

Generative AI holds great promise for reshaping supply chains, exemplified by Zara's 10 million dollars savings in 2020, but it also raises concerns about job displacement. The shift toward AI necessitates extensive reskilling programs, which require investments in workforce adaptation [54], [107]. Quantum computing, while offering an advantage in optimizing logistics and reducing delays by 15%, also introduces cybersecurity risks, as it may expose sensitive data to potential breaches [55]. These technologies, though beneficial, require careful management of their dual nature, balancing their positive outcomes with the risks they introduce, to strengthen OSCM systems and mitigate economic volatility during macroeconomic crises.

Policymakers play a critical role in supporting these efforts, balancing innovation with stability across disruptors. Incentives for green technology adoption, such as tax credits covering 30% of renewable energy costs, could offset blockchain and AI's energy demands, saving organizations 2–5 billion dollars collectively by 2030 [41]. Regulatory frameworks for quantum-safe cybersecurity, costing 500 dollars million globally to implement, protect against encryption risks during trade tensions or pandemics [55]. Funding labor transition programs, estimated at 1 trillion dollars globally, mitigates AI-driven job losses, ensuring social stability during regulatory or health crises [107]. These policies, tailored to trade, regulatory, pandemic, and energy contexts, foster an ecosystem where OSCM thrives amidst uncertainty.

Future research directions for OSCM scholars should explore these technologies' systemic impacts, building on their applications and implications without redundancy. Table III provides a structured summary of the future research directions, visually mapping how each technology addresses key thematic areas. First, investigating energy tradeoffs requires quantitative models; assessing blockchain's PoS scalability could use simulation studies to predict a 50% cost reduction in energy-intensive supply chains by 2025 [52]. Second, labor market dynamics demand longitudinal analyzes; tracking generative AI's 75 million job displacement forecast through case studies in pandemic-affected regions could identify reskilling, targeting a 70% reemployment rate [107]. Third, governance frameworks for quantum computing's cybersecurity risks need policy simulations; modeling a 1 billion dollars global encryption upgrade could ensure trade tension resilience by 2030 [55]. Fourth, systemic interactions among technologies and disruptors warrant agent-based modeling; a 2025 study predicting a 20% resilience boost from combined blockchain-digital twin use could guide OSCM strategies [53]. These avenues, grounded in data-driven methodologies, deepen understanding of how these technologies interplay with trade tensions, regulatory changes, pandemics, and energy issues, offering actionable insights for practitioners and policymakers.

Furthermore, a future research direction could involve the development of knowledge management systems (KMS) that are designed to operate within the emerging multilayered information architecture of OSCM. As blockchain, digital twins, generative AI, and quantum computing generate, process, and analyze vast amounts of structured and unstructured data, there is a need for knowledge management frameworks that can capture, organize, and operationalize this information to mitigate risks associated with macroeconomic disruptions. Unlike traditional KMS, which focus on static documentation and expert-driven knowledge curation, next-generation systems must dynamically integrate real-time insights, simulations, and predictive models into decision workflows. This calls for rethinking how tacit and explicit knowledge is identified, codified, and disseminated across distributed and autonomous supply chains and project management information systems [108], [109], [110]. Future studies could explore how to architect such KMS to support adaptive learning, scenario-based reasoning, and decentralized knowledge validation.

Moreover, an emerging research direction involves exploring the early-stage development of omnitrust and autonomous

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supply chains that are decentralized and intelligent, capable of operating with reduced reliance on traditional forms of interpersonal and interorganizational trust. However, the practical implementation of such systems remains limited, and many questions persist regarding their feasibility, governance, and broader implications. Precisely, the shift toward omnitrust raises new challenges related to metadata security, economic accessibility, and sustainability. Future research should understand the conditions under which omnitrust-based architectures can evolve from conceptual promise to operational reality will be essential for advancing resilient supply chains.

Finally, a promising research direction lies in examining how the adoption of advanced technologies in OSCM should be studied in parallel with geopolitical and macroeconomic dynamics such as partial deglobalization and trade tensions. These forces are reshaping global supply chains, not only through regulatory and economic barriers but also through potential shifts in territorial control and resource accessibility. For example, geopolitical ambitions regarding Arctic territories, combined with environmental changes such as Arctic ice melt, could open new transportation corridors and expose untapped reserves of critical underground resources. These developments suggest that supply chains must anticipate and model the implications of emerging trade routes, territorial claims, and resource discoveries. Studying the intersection of advanced technologies with these macrostructural disruptions could redefine how supply chains plan distribution, sourcing, and infrastructure investments in a reglobalizing and climate-impacted world.

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Jiho Yoon received the Ph.D. degree in supply chain management from Michigan State University, East Lansing, MI, USA, in 2015.

He is currently an Associate Professor of Management Science with Chung-Ang University, Seoul, South Korea, where he is currently the Director of the CAU Internship Support Center (CAU ISC). Previously, he was an Assistant Professor with Kansas State University, Manhattan, KS, USA, and taught at Worcester Polytechnic Institute, Worcester, MA, USA. His research focuses on the areas of risk management, energy issues, sustainability, and advanced technologies (AI, Blockchain, Digital twin, etc.) implementation in logistics and supply chains. His research has appeared in the *Decision Sciences*, *Journal of Business Logistics*, *International Journal of Production Research*, *Journal of Business Research*, *Journal of the Operational Research Society*, IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT, *OMEGA*, among others.



Rami Alkhudary received the bachelor's degree in economics from Damascus University, Damascus, Syria, in 2011 and the master's degree in international management from IAE Poitiers, Poitiers, France, in 2014, and the Ph.D. degree in management science, with a focus on blockchain applications in supply chain management, in 2021, and the Habilitation à Diriger des Recherches (HDR) in management science from Paris Panthéon-Assas University, Paris, France, in 2025.

He is currently an Associate Professor of Supply Chain Management and Information Systems with Paris Panthéon-Assas University, where he codirects several academic programs and is responsible for the distribution, logistics, industrial marketing, and supply chain management tracks in two bachelor's degrees and one master's program, cofiled a U.S. technology-based patent in collaboration with McGill University. His research focuses on the use of emerging technologies in supply chain and project management. His work has been published in journals such as the *International Journal of Production Research*, *International Journal of Project Management*, *Marketing Letters*, *European Business Review*, *The International Journal of Logistics Management*, and *Harvard Business Review France*, among others.

Dr. Alkhudary is currently a Reviewer for several academic journals and conferences. He was the recipient of the research grants from the Project Management Institute (USA) and Fundação Getulio Vargas (Brazil).

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Srinivas Talluri received the Ph.D. degree in production and operations management from the University of Texas, Arlington, TX, USA, in 1996. He is currently the Hoagland Metzler Endowed Professor and a Professor of Supply Chain Management with Michigan State University, East Lansing, MI, USA. His research interests are in the areas of supply risk, buyer–supplier relationships, sustainability in supply chains, and performance evaluation. His methodological expertise includes deterministic and stochastic modeling, game theory, empirical methods, and data envelopment analysis. He has authored or coauthored more than 100 articles in journals, such as *IIE Transactions*, *Journal of Operations Management*, *Decision Sciences*, *Production and Operations Management Journal*, *International Journal of Production Research*, *European Journal of Operational Research*, IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT, *Journal of Business Logistics*, and others. His research has garnered more than 16 500 citations with an H-index of 62.

Mr. Talluri is currently the Co-Editor in Chief of the *Decision Sciences Journal*. He was the Department Editor for IEEE TRANSACTIONS ON ENGINEERING MANAGEMENT. He is a Fellow of the Decision Sciences Institute, and a member of the Decision Sciences Institute and the Production and Operations Management Society.



Pierre Féniès received the Ph.D. degree in information technology with a focus on the modeling of industrial performance and supply chain management, from Université Clermont Auvergne, Clermont-Ferrand, France, in 2006.

He is currently a Full Professor of Logistics and Supply Chain Management with Paris Panthéon-Assas University, Paris, France. He is currently the Dean of the Faculty of Management Science. He also directs the Master's program in Project Logistics Management and oversees the university's logistics track (filière Logistique). His areas of expertise include management control, logistics, supply chain management, and business intelligence. He has supervised tens of Ph.D. candidates. His research and teaching highlight the importance of integrating IT with decision-making in management control to enhance software solutions and organizational processes, both in practice and in research. His research has been validated in several industrial and hospital supply chains. He has collaborated with IP Leanware on the development of business intelligence software for industrial performance management. His work has been authored or coauthored in journals such as the *International Journal of Production Economics*, *International Journal of Production Research*, *Computers and Industrial Engineering*, and *Production Planning and Control*, among others.

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